

Local data

Predictors

Research loop

Aggregation

OUTER RESEARCH LOOP

Research language model: fixed weights

Evidence cards + previous trials

1 Propose

Hypothesis +
design identifier

2 Instantiate

Resolve full recipe;
build executable
candidate

3 Train + evaluate

Fit locally; return
scores + diagnostics

4 Retain / revise

Higher mean dev score;
tie → lower mean loss

Candidate recipe (full
config)

Dev evidence:
score + loss +
diagnostics

Selected

INNER COLLABORATIVE TRAINING

Shared coordinator: run, aggregate, select

Broadcast parameters; aggregate local states + development scores

Lab 1

Scenario A

Train / dev

Local model

Lab 2

Scenario B

Train / dev

Local model

Lab K

Scenario K

Train / dev

Local model

Measurements remain within each laboratory

OUTPUTS

Selected design
+ fitted predictor

Research history
all trial results

TASK-SPECIFIC PREDICTORS AND SCORERS

Drug-target interactions

Compound + protein to interaction

Proteomic efficacy

Observed proteome + drug to efficacy

Cell perturbations

Single gene / double gene / drug